



Computational Modelling and Simulation M.Sc.
Life Science Track

Research Project Report:

Examiner Coach

A Modular RAG Prototype for OSCE Examiner Feedback Training

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Abstract

This report presents Examiner Coach, a prototype system for supporting OSCE examiner feedback training through speech transcription, structured feedback-quality evaluation, and retrieval-augmented coaching. The system models examiner feedback quality using six formative criteria and implements a backend-centered modular retrieval-augmented generation pipeline for evidence-grounded evaluation and follow-up support. The project is motivated by a training gap in OSCE examinations: examiners are expected to deliver concise, structured feedback, but often lack scalable opportunities for individual practice, immediate correction, and privacy-conscious AI support. Inspired by Modular RAG concepts, the implemented pipeline separates knowledge indexing, transcript normalization, retrieval, post-retrieval evidence selection, LLM-based generation, and validation. A prototype comparison of RAG variants illustrates the effect of retrieval strategies, including direct retrieval and HyDE-based retrieval, on calibration against benchmark feedback cases. The report focuses on the backend modelling and retrieval architecture, while the user interface is treated only as a supporting layer for audio input and result display.

1 Introduction

1.1 Motivation

Objective Structured Clinical Examinations (OSCEs) are widely used in medical education to assess clinical and communication skills. In this setting, the quality of examiner feedback is central to learning. However, feedback may vary substantially between examiners and can be vague, overly judgmental, insufficiently specific, or difficult for students to act upon. Feedback that does not refer to observable behavior can support encouragement, but it often does not provide enough information for targeted improvement [KB3, KB2, KB10].

This problem is particularly relevant in current examiner training. Examiners are required to give structured two-minute feedback at every OSCE station, but many have not received formal training in how to do this. The skill is not trivial: high-quality feedback requires observation, prioritisation, concise wording, balanced judgement, and an actionable recommendation within a short time window. An examiner may know the expected clinical performance, but still struggle to formulate feedback that is educationally useful.

For example, a statement such as “good job” or “you need to be more systematic” may be well intentioned, but it does not tell the learner which observed action was effective, which behavior should be changed, or how to improve in the next OSCE station. Examiner Coach addresses this gap by treating feedback itself as the object of analysis. The system does not grade the student in the OSCE video; instead, it evaluates the quality of the examiner’s spoken feedback to that student. The project therefore sits at the intersection of medical education, natural language processing, and human-computer interaction.

1.2 Problem Definition

The problem addressed in this report is not simply that feedback can be poor. It is that feedback delivery is a skill that requires practice, guidance, and timely correction, while the current OSCE training context provides limited support for individual rehearsal. The project is defined by four connected challenges:

- examiners are expected to provide structured two-minute feedback at every station, but many have received no formal feedback-delivery training;
- existing examiner training is often delivered in large-group workshops with limited individual practice and personalised correction;
- there is no scalable, on-demand tool that allows examiners to practise feedback delivery and receive immediate criterion-based evaluation;
- privacy constraints in medical education make commercial AI services unsuitable for many examiner-training scenarios, because patient-adjacent and institutional data must remain protected.

The resulting training gap is therefore clear: examiners are expected to give structured, useful feedback, but lack a scalable and privacy-conscious way to rehearse this skill before entering the examination setting.

1.3 Aim and Contributions

This project aims to develop a prototype AI system that supports examiner feedback training by evaluating spoken feedback and providing formative coaching. The main contributions are:

- a structured six-criterion model for feedback-quality evaluation;
- a backend-centered modular RAG pipeline for evidence-grounded assessment;

- bilingual German and English evaluation and coaching support;
- a prototype comparison of RAG variants, including no-RAG, direct retrieval, and HyDE-based retrieval.

The report therefore asks how a modular RAG architecture can be adapted to a medical education feedback-training scenario, and whether retrieval-based variants provide clearer calibration than a prompt-only baseline. The intended outcome is not a final clinical validation, but a technically inspectable prototype that can support later expert review and user studies.

2 Background

2.1 Feedback Quality in Medical Education

Effective feedback in clinical education should be specific, behavior-oriented, objective, and actionable. It should identify both strengths and areas for improvement while avoiding vague praise or personal judgment. In OSCE examiner training, these requirements are especially important because examiners must provide concise, educationally useful feedback based on observed student performance.

The distinction between feedback, encouragement, and evaluation is important for this project. Encouragement may support motivation, but it often lacks information value. Evaluation can summarize performance, but may remain too summative to guide improvement. Formative feedback should instead describe a performance gap and help the learner close it. In an OSCE setting, this means that useful examiner feedback should name what the student did, connect the comment to the station context, use descriptive rather than personal language, and provide a concrete strategy for the next attempt.

These requirements become harder under the practical constraint of a two-minute feedback window. The examiner must select the most relevant points, formulate a strength, identify an improvement area, and give an actionable next step without becoming either too vague or too lengthy. This makes feedback delivery a trainable performance skill in its own right. A scalable training tool should therefore allow examiners to rehearse feedback repeatedly and receive immediate, criterion-based guidance.

This educational view motivates the six-criterion rubric used in Examiner Coach. The rubric does not attempt to model all aspects of clinical competence. It models the quality of the examiner’s feedback utterance as a training object. This separation is essential because the same student performance could receive feedback of very different quality depending on the examiner’s wording.

2.2 Retrieval-Augmented Generation and Modular RAG

Retrieval-augmented generation combines large language models with external knowledge retrieval. Instead of relying only on model-internal knowledge, a RAG system retrieves relevant evidence and provides it as context for generation. The Modular RAG framework describes RAG systems as configurable combinations of modules such as indexing, pre-retrieval, retrieval, post-retrieval, generation, and orchestration [T1]. Examiner Coach follows this idea by separating its knowledge preparation, transcript processing, retrieval, scoring, and coaching components.

The Modular RAG perspective is useful because Examiner Coach is not simply a single prompt around a transcript. It is a pipeline with several points at which behavior can be changed and evaluated independently: documents can be chunked differently, retrieval can use direct transcript queries or HyDE-style hypothetical-passage retrieval, retrieved chunks can be filtered or reranked, and generation can be constrained by a structured JSON contract. This modularity improves inspectability. When an evaluation seems poorly calibrated, the system can be debugged at the level of retrieval evidence, prompt construction, parsing, or rubric design rather than being treated as an opaque LLM response.

2.3 KISSKI Context for Sensitive Health Applications

The project also fits into the broader direction of KISSKI, the AI service centre for sensitive and critical infrastructures. KISSKI’s central goal is to research AI methods and provide them

through a highly available AI service centre for sensitive infrastructures, with a particular focus on socially relevant fields such as health and energy [T8]. Examiner Coach takes a step in this direction by applying institutionally provided AI services to a concrete health-education use case: OSCE examiner feedback training. The project is not only a consumer of an LLM endpoint; it explores how such services can be integrated into a structured, privacy-conscious workflow for medical education.

The modular perspective also supports the privacy-oriented requirements of the project. Instead of sending examiner input to a general commercial chatbot, the system is configured around institutional KISSKI/SAIA services and a local knowledge base. This does not remove all privacy concerns, but it makes the data flow more explicit and gives the project a clearer path toward institutional review.

3 System Design

3.1 Backend-Centered Pipeline

The backend AI pipeline represents the core of Examiner Coach. The pipeline receives spoken feedback, obtains a transcript, retrieves relevant educational evidence, evaluates the transcript against a rubric, and supports follow-up coaching.

The pipeline can be summarized as follows:

1. the examiner records spoken feedback for an OSCE station;
2. the audio is transcribed by the configured speech model;
3. the transcript is normalized where needed for retrieval;
4. criterion-aware retrieval queries are constructed;
5. educational feedback literature is retrieved from the vector store;
6. the LLM evaluates the transcript using the rubric and retrieved evidence;
7. the structured result is validated and resolved into the requested display language;
8. optional follow-up coaching chat reuses the transcript, evaluation, and retrieved evidence.

The frontend is therefore intentionally not the scientific focus of this report. It acts as an interaction layer for recording and displaying results, whereas the research contribution is the modelling and orchestration of the backend evaluation and coaching process.

The design is shaped by the problem definition: the system must be available for individual practice, return feedback immediately enough to support rehearsal, and evaluate the examiner’s response against explicit criteria rather than providing only generic comments. It must also keep the data flow inspectable so that privacy implications can be discussed and controlled in an institutional medical education context. In this sense, the project can be understood as a small applied step toward the KISSKI vision: providing AI methods for sensitive health-related infrastructures in a way that is usable for real institutional workflows rather than only for isolated demonstrations.

3.2 Knowledge Base and Retrieval

Educational feedback material is ingested into a local ChromaDB vector store. Documents are chunked, embedded using a multilingual embedding model respecting the bilingual nature of both the knowledge base and the feedback transcripts, and retrieved at runtime based on the feedback transcript and evaluation criteria. Retrieval can be performed directly or through HyDE-based hypothetical-passage retrieval.

The knowledge base gives the system a way to ground feedback evaluation in educational guidance instead of relying only on the LLM’s internal training data. This is especially important because the task is not generic text classification. The system must distinguish feedback that is merely positive from feedback that is specific, formative, and behaviorally actionable. The retrieved context provides concepts and examples that help the evaluation prompt remain aligned with feedback education principles.

3.3 Evaluation and Coaching Flow

The evaluation flow combines the transcript, retrieved evidence, and a structured rubric prompt. The LLM returns a validated structured response containing criterion-level scores, suggestions, quotes where applicable, and an overall score. The coaching flow reuses the transcript, evaluation result, and retrieved evidence to answer follow-up questions.

Module	Implementation in Examiner Coach	Purpose
Indexing	Document ingestion, chunking, embeddings, ChromaDB storage	Prepare educational evidence for later retrieval.
Pre-retrieval	Transcript normalization and criterion-aware query construction	Improve retrieval relevance across German and English feedback.
Retrieval	Direct vector search or HyDE-based retrieval	Select candidate evidence chunks from the knowledge base.
Post-retrieval	Filtering, reranking, and final evidence formatting	Prefer practical feedback guidance over generic or weakly related chunks.
Generation	Structured LLM prompt using transcript, evidence, and rubric	Produce criterion scores, suggestions, quotes, and summary.
Validation	JSON parsing and schema-based result assembly	Keep output machine-readable and suitable for coaching reuse.

Table 1: Mapping of Examiner Coach to a modular RAG architecture.

The coaching component extends the evaluation from a static score report into a learning interaction. After receiving an evaluation, the examiner can ask why a criterion was scored in a particular way or request alternative wording. This is important pedagogically because the score alone does not teach the examiner how to improve. The coaching response can refer back to the transcript, the scored criteria, and the evidence base, thereby connecting system output to concrete feedback practice.

4 Feedback Quality Model

4.1 Six-Criterion Rubric

Examiner Coach evaluates feedback using six criteria:

1. specific observed behavior named;
2. contextual feedback;
3. objective and non-evaluative tone;
4. explicitly mentioned strength;
5. changeable area for improvement;
6. concrete improvement plan or next step.

These criteria were selected to represent feedback as an educationally useful communication act. The first criterion asks whether the examiner names a specific observed behavior, because learners can only improve reliably when they know which action is being discussed. The second criterion checks whether the comment is connected to the current OSCE station rather than being a generic statement that could apply to any student. The third criterion addresses tone: the system rewards descriptive, behavior-focused wording and penalizes unsupported personal judgment.

The fourth and fifth criteria ensure that the feedback is balanced but not superficial. A strength should be explicitly named, and an improvement area should describe a changeable behavior rather than a fixed personal trait. The sixth criterion captures actionability. It is not enough to tell a student that something was missing; high-quality feedback should include a concrete next step, such as a structure to follow, a phrase to use, or a specific clinical sequence to practice.

4.2 Scoring and Bilingual Output

Each criterion is scored from 0 to 100 using predefined scoring anchors. The system computes an overall score as the average of the six criterion scores and counts criteria as met when they meet or exceed the default threshold (≥ 70). Evaluation and coaching results can be resolved into German or English without regenerating the whole evaluation.

The scoring model is intentionally simple. It does not claim to measure feedback quality as an objective natural quantity; instead, it operationalizes feedback quality for formative training. A score below 70 signals that a criterion is absent or only weakly expressed, while a score of 70 or above indicates that the criterion is present at a usable level. This makes the result easy to interpret and allows the coaching component to focus on the weakest criteria.

Bilingual output is relevant because OSCE examiner training at TU Dresden may involve German-language feedback while the technical and educational literature is often available in English as in our case. The backend keeps evaluation content in a structured form and resolves labels, summaries, and suggestions into the requested language. This avoids regenerating the whole evaluation solely to switch the display language.

5 Implementation

5.1 Transcription, RAG, and LLM Evaluation

The backend is implemented as a FastAPI service and is organized around a small set of domain-specific services. This design keeps the main AI workflow separate from the user interface and makes each processing step easier to test and inspect. The runtime pipeline begins with an audio recording of examiner feedback. The transcription service sends this audio to the configured KISSKI/SAIA speech endpoint and returns the transcribed text together with the audio duration.

After transcription, the RAG pipeline prepares the text for evidence-grounded evaluation. If required, the transcript is normalized into English so that retrieval can work consistently across German and English inputs. The pipeline then builds a criterion-aware retrieval query, retrieves relevant educational evidence from the ChromaDB knowledge base, and applies filtering and quality reranking to select the final evidence chunks. These chunks are not used as a replacement for the transcript; rather, they provide educational guidance that helps the LLM evaluate the feedback according to the defined rubric.

The knowledge base is prepared separately by the document-management service. Source documents are parsed, split into semantically useful chunks, embedded with the configured multilingual embedding model, and stored in ChromaDB. This separation between offline knowledge-base preparation and online evaluation makes the system easier to maintain. For example, a transcription failure, retrieval problem, prompt issue, or JSON parsing error can be localized to a specific service rather than being hidden inside one monolithic workflow.

The evaluation prompt combines three elements: the transcript, the retrieved evidence, and the six-criterion feedback-quality rubric. It also defines the expected response format. The LLM is asked to return a structured object containing a short summary, criterion-level scores, suggestions, a key suggestion, and optional quotes from the transcript. The backend then attaches the transcript and duration, computes the overall score and the number of met criteria, validates the result, and resolves it into German or English. This structured response is important because it can be compared across RAG variants and reused by the coaching endpoint. A purely free-form answer would be harder to parse, evaluate, and connect to later system components.

The implementation also includes an interactive coaching step. Unlike the initial evaluation prompt, the coaching prompt receives the previous evaluation, the original transcript, the user’s follow-up question, recent conversation history, and retrieved evidence when relevant. This allows the system to explain why a criterion received a certain score, suggest clearer feedback wording, or connect an improvement suggestion to educational guidance. In this way, the system moves beyond a one-time automated score and becomes a practice-oriented feedback training tool.

5.2 Model Configuration and Reproducibility

The current system configuration uses:

- evaluation model: `meta-llama-3.1-8b-instruct`;
- judge model: `gpt-oss-120b`;
- embedding model: `multilingual-e5-large-instruct`;
- transcription model: `whisper-large-v2`.

The current production RAG configuration follows the strongest variant observed in the

comparison workflow: `hyde_k8`. Both evaluation and coaching retrieval use HyDE-based retrieval with a candidate pool of 20, a final context size of 8 chunks, English normalization enabled, criterion-aware query construction enabled, and quality reranking enabled. In this setting, the pool is split between direct retrieval from the transcript and retrieval from a HyDE-generated hypothetical guidance passage. The resulting candidates are deduplicated, reranked, filtered, and reduced to the final evidence context passed into the LLM.

The RAG comparison outputs in the debug notebook were generated when the evaluation model was configured as `llama-3.3-70b-instruct`. That model is no longer supported in the current KISSKI setup, so the runtime evaluation model was changed to `meta-llama-3.1-8b-instruct`. The judge model remains `gpt-oss-120b`, which is still the configured judge model.

This distinction matters for reproducibility. The notebook results should be understood as an illustration of the retrieval design and comparison method rather than a fully reproducible final benchmark under the current model configuration. Future evaluation should repeat the comparison with the currently available evaluation model and report whether the same retrieval ranking holds.

5.3 Implementation Flowchart

Figure 1 summarizes the implemented backend workflow. The diagram separates the offline knowledge-base preparation from the runtime examiner-feedback pipeline. This distinction is important because the knowledge base can be updated independently, while the runtime path is executed for each examiner feedback submission or coaching question.

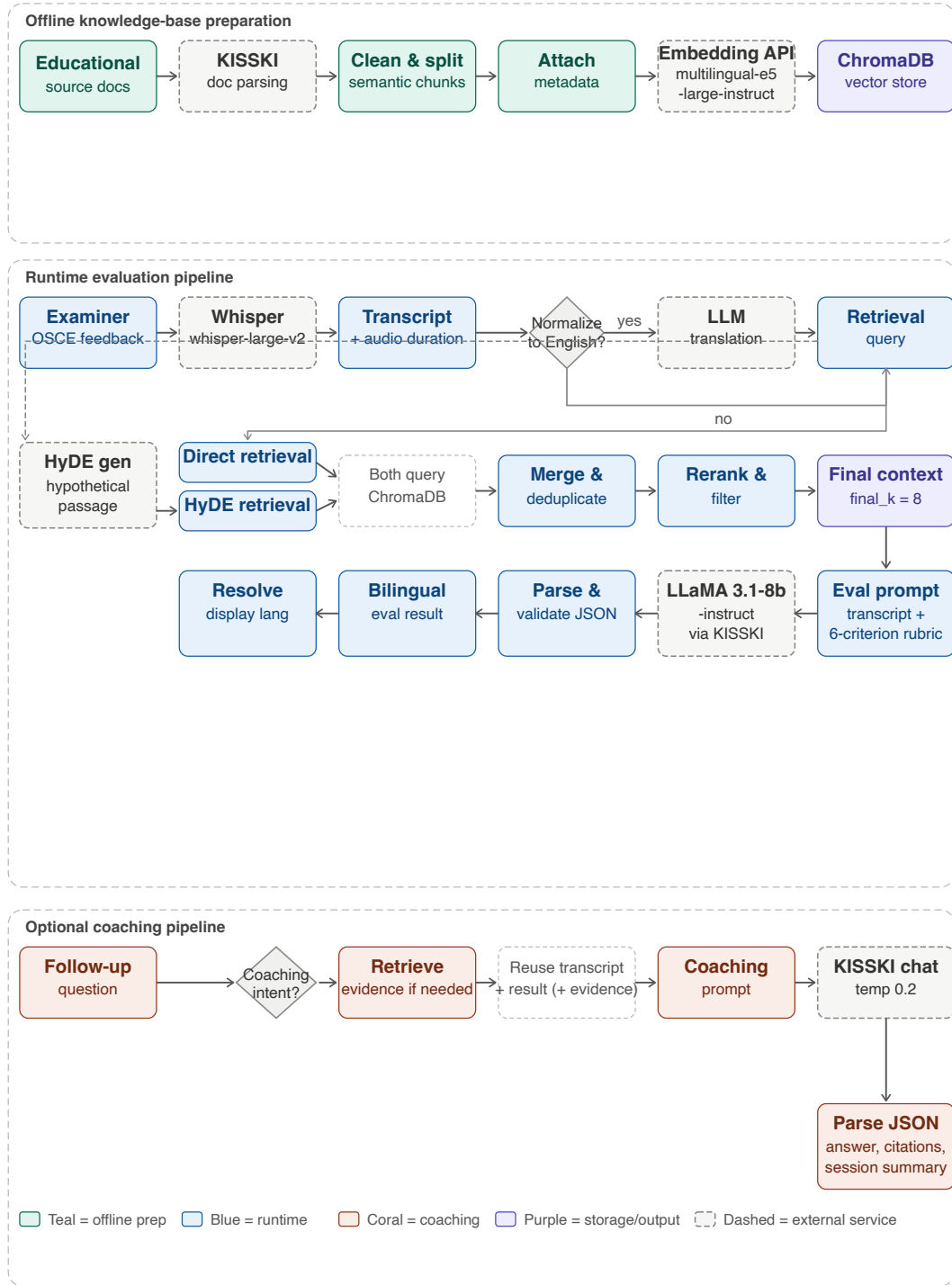


Figure 1: Detailed implementation flow of Examiner Coach, including knowledge-base preparation, runtime evaluation, RAG retrieval, and optional coaching.

6 Prototype Evaluation

6.1 RAG Variant Comparison

Prototype evaluation was performed using benchmark feedback cases and several retrieval configurations:

- `no_rag`;
- `direct_k8`;
- `direct_k8_unfiltered`;
- `hyde_k8`;
- `hyde_k8_unfiltered`.

The outputs from `notebooks/rag_debug_demo.ipynb` illustrate how retrieval affects score calibration and feedback interpretation.

The comparison is designed to answer a practical calibration question: does retrieval help the system place feedback examples closer to expected quality bands? The no-RAG variant provides a prompt-only baseline. Direct retrieval tests whether evidence from the knowledge base improves evaluation when the retrieval query is based directly on the transcript and the feedback-quality criteria. HyDE retrieval adds one additional step: the LLM first generates a short hypothetical guidance passage that resembles the kind of educational evidence that would be useful for judging the transcript. This generated passage is then embedded and used to retrieve knowledge-base chunks. In other words, direct retrieval searches with the evaluation query itself, whereas HyDE retrieval searches with a generated document-like representation of the evidence that the system hopes to find.

The comparison also separates filtered and unfiltered retrieval. In the unfiltered variants, criterion-aware query construction and quality reranking are disabled, so retrieval remains closer to the raw transcript-based vector similarity ranking:

```
query -> retrieve candidates -> sort by relevance -> take final_k chunks
```

The filtered variants add two refinements: the retrieval query is expanded with the feedback-quality criteria, and post-retrieval quality control is applied before the final context is selected:

```
query -> retrieve candidates -> quality reranking -> remove low-value chunks ->
take final_k chunks
```

This distinction is important because vector similarity alone can retrieve chunks that are semantically related but not educationally useful, such as generic introductions, conclusions, reference sections, or noisy document fragments. The criterion-aware query pushes retrieval toward guidance about the six rubric dimensions, while quality reranking gives higher priority to chunks that mention specific observed behavior, objective wording, changeable improvement areas, and concrete next steps. The unfiltered variants therefore answer the question of how far raw retrieval can go, while the filtered variants test whether task-specific query construction and evidence-selection logic improve calibration.

The benchmark cases are grouped by expected quality bands such as weak, mediocre, strong, and edge cases. This grouping is useful because a model may perform well on obviously strong feedback while still struggling with borderline or moderately useful feedback. For formative examiner training, those middle and edge cases are especially important because they resemble realistic feedback that is partially useful but still improvable.

Variant	Avg. distance to gold band	Within band	Interpretation
hyde_k8	2.19	10/16	Best overall alignment
hyde_k8_unfiltered	2.31	9/16	Very close to filtered HyDE
direct_k8	2.56	8/16	Useful, but weaker than HyDE
direct_k8_unfiltered	3.00	6/16	Lower calibration
no_rag	3.31	7/16	Prompt-only baseline

Table 2: Illustrative RAG comparison from the debug notebook. Lower distance to the gold band is better.

6.2 LLM-as-a-Judge Validation

In addition to the deterministic distance-to-gold-band comparison, an LLM-as-a-judge step was used as a secondary calibration check. The judge read the already generated evaluation outputs from the RAG comparison and compared them with the benchmark expectations. In the debug notebook, this produced 80 completed judgments, corresponding to 16 benchmark cases evaluated under five variants.

The judge was not treated as ground truth. Instead, it acted as a structured second lens beside the numeric band-distance calculation. This is important because an overall score can be numerically close to the target band while the reasoning remains educationally weak, or a criterion score can appear plausible while the explanation does not match the intended benchmark expectation. The judge therefore assessed four complementary dimensions:

- **Overall band fit:** whether the final feedback-quality score places the examiner’s feedback in the expected benchmark category, such as weak, mediocre, strong, or edge-case feedback. This checks the global calibration of the evaluation.
- **Criterion band fit:** whether the six rubric criteria are scored consistently with the benchmark expectation. For example, feedback that contains no concrete next step should not receive a high score for the improvement-plan criterion, even if its general tone is acceptable.
- **Benchmark expectation match:** whether the generated summary, reasoning, and suggestions explain the same strengths and weaknesses that the benchmark case was designed to test. This checks whether the system understood why a feedback example should be considered weak, partial, or strong.
- **Directional correctness:** whether the evaluation moves in the correct formative direction. In this project, that means avoiding inflated scores for vague feedback, avoiding excessive penalties for clearly structured feedback, and preserving the intended order between weak, mediocre, strong, and ambiguous examples.

Variant	Mean judge score	Overall band fit	Criterion band fit	Benchmark match	Directional correctness
hyde_k8	62.97	71.25	52.50	51.56	76.56
hyde_k8_unfiltered	62.61	67.19	55.88	54.88	72.50
direct_k8	55.88	67.50	46.94	42.50	66.56
direct_k8_unfiltered	54.61	53.12	49.81	49.88	65.62
no_rag	52.70	54.38	48.31	47.19	60.94

Table 3: LLM-as-a-judge summary from the debug notebook. Scores are reported on a 0–100 scale; higher values indicate stronger agreement with the benchmark expectation.

The judge results support the same overall direction as the deterministic comparison in Table 2. The strongest mean judge score belongs to the filtered HyDE variant, with the unfiltered HyDE variant following very closely. This indicates that the advantage of HyDE is not only visible in the numeric distance to the gold band, but also in the judged quality of the generated evaluation. The filtered HyDE variant achieved the highest overall band fit and directional correctness, suggesting that it was best at placing examples in the right broad quality region and preserving the expected weak/mediocre/strong/edge direction.

The unfiltered HyDE variant was very close and performed slightly better on criterion band fit and benchmark expectation match. This suggests that raw HyDE retrieval may sometimes preserve criterion-level details that are useful for the judge. However, the filtered `hyde_k8` variant remained stronger as an operational default because it gave the best balance between overall calibration, directional correctness, and robustness against low-value context.

6.3 Interpretation of Results

Taken together, the deterministic and judge-based results show a consistent pattern. Retrieval improves the evaluation when it provides evidence that is well aligned with the feedback-quality rubric, and HyDE appears to improve this alignment more reliably than direct retrieval alone. The selected `hyde_k8` configuration achieved the lowest average distance to the gold band, the highest number of in-band cases, the highest mean judge score, and the strongest directional correctness. This made it the most convincing default configuration for the current prototype.

The comparison also shows that the no-RAG baseline is not useless. A carefully designed rubric prompt already gives the LLM meaningful structure, and this is why the prompt-only baseline can still produce reasonable evaluations in some cases. The added value of RAG is therefore not that it creates the rubric from nothing, but that it improves grounding and calibration by placing the transcript beside relevant educational evidence.

The difference between filtered and unfiltered retrieval should be interpreted carefully. Filtering and quality reranking help remove generic or noisy chunks and prioritize evidence about behavior-specific, objective, and actionable feedback. At the same time, the close performance of `hyde_k8_unfiltered` shows that the generated HyDE passage itself already retrieves useful evidence. This means that both components matter: HyDE improves the semantic match between the evaluation task and the knowledge base, while filtering tries to make the final context more reliable.

The most difficult category remains moderately useful feedback. Strong examples are easier to recognize because they satisfy most criteria, and weak examples are easier because many criteria are absent. Mediocre feedback often contains a mixture of useful and missing elements, making calibration more sensitive to retrieved evidence, scoring anchors, and prompt wording. This observation is important for future validation because real examiner feedback is likely to fall frequently into this intermediate range. Future work should therefore repeat the comparison with a larger benchmark set and with the currently available KISSKI evaluation model to test whether the same ranking remains stable.

7 Discussion

7.1 Strengths

The main strength of Examiner Coach is its modular backend design. The system separates transcription, retrieval, evaluation, validation, and coaching, making it easier to inspect and adapt individual components. The six-criterion rubric also makes LLM output more structured and educationally interpretable.

Another strength is the direct connection between educational theory and system design. The system does not merely ask an LLM to “give feedback on feedback”. It constrains the task through explicit criteria, score anchors, retrieved evidence, and a structured output schema. This makes the output more suitable for formative training because the examiner can see which aspect of the feedback needs improvement and can ask targeted follow-up questions.

This directly addresses the training gap described in the problem definition. Instead of relying only on occasional large-group examiner training, the prototype supports repeated individual practice: an examiner can record a two-minute feedback attempt, receive immediate criterion-based evaluation, and then ask for clarification or alternative wording. The system therefore targets the missing practice-and-correction loop rather than replacing formal examiner training.

The modular design also supports experimentation. Retrieval can be disabled, direct retrieval and HyDE can be compared, and filtering or reranking can be switched on or off. This is valuable in a research project because it allows the system to be studied as a set of computational modelling choices rather than as a fixed black-box application.

7.2 Limitations

The current evaluation is prototype-level and should not be interpreted as a final validation study. Results depend on transcription quality, retrieval quality, prompt calibration, model availability, and the benchmark design. The system does not replace expert human judgment.

The benchmark comparison is also limited by its size and construction. Expected score bands provide a useful calibration target, but they are not equivalent to consensus ratings from multiple medical education experts. The LLM-as-a-judge analysis gives an additional consistency check, but it remains model-based and therefore cannot replace human validation. A future study should compare system outputs against trained human raters and examine inter-rater agreement.

Another limitation is model stability. The evaluation model used for the notebook comparison is no longer available in the current KISSKI setup, so the exact results should not be overinterpreted as final performance claims. The architecture and comparison method remain valid, but the numerical results need to be repeated under the current model configuration.

7.3 Privacy and Ethical Considerations

The system processes audio, transcripts, evaluation results, and coaching interactions. External AI services may receive audio or text depending on the configured endpoint. Future deployment requires clear retention policies, access control, participant information, and review of institutional data protection requirements.

This privacy aspect is part of the core problem, not an afterthought. In medical education, examiner feedback may be patient-adjacent, institution-specific, or linked to assessment material. For this reason, a tool intended for examiner training cannot simply assume that commercial AI services are acceptable. The prototype uses configured KISSKI/SAIA endpoints and a local knowledge base, but future deployment would still need formal review of what data leaves the

local environment, how long audio and transcripts are retained, and who can access the stored training outputs.

Ethically, the system should be positioned as a training aid rather than an authority over examiner competence. Automated feedback evaluation can support self-reflection, but it may also create false confidence if users treat scores as objective judgments. For this reason, the report interprets scores as formative indicators and emphasizes the need for expert review before live study deployment.

7.4 Future Work

Future work should first validate the prototype with medical educators and OSCE examiners. The current benchmark provides an initial calibration signal, but it should be compared with human expert ratings and tested in a user study that examines usefulness, trust, clarity of the criteria, and whether repeated practice improves feedback delivery.

Technically, the next step is more detailed retrieval analysis. This includes examining which evidence sources are retrieved, which rubric criteria benefit most from RAG, and how retrieval behaves across German and English transcripts. The comparison framework could also be extended to test different values of `final_k`, candidate-pool sizes, HyDE prompts, reranking rules, and repeated model runs.

The extended comparison was not used in this report to avoid unnecessary use of shared AI infrastructure and to remain mindful of the environmental impact of large numbers of LLM and judge calls. Future experiments should therefore be planned around specific research questions and reported together with the model, number of calls, and configuration used.

8 Conclusion

Examiner Coach demonstrates how a modular RAG architecture can support formative OSCE examiner feedback training. By combining speech transcription, structured feedback-quality modelling, evidence retrieval, and LLM-based coaching, the system provides a practical prototype for AI-assisted medical education. The prototype evaluation suggests that retrieval, especially HyDE-based retrieval, can improve calibration compared with a no-RAG baseline, although further expert validation is required.

The project’s main value lies in making feedback quality computationally inspectable without reducing it to a single opaque score. The six-criterion rubric gives examiners concrete dimensions for reflection, while the modular RAG pipeline provides a way to ground evaluation and coaching in educational evidence. As a research prototype, Examiner Coach therefore offers both a technical contribution to applied RAG design and a practical contribution to medical education feedback training.

In practical terms, the prototype responds to a concrete institutional training need: examiners are expected to deliver structured feedback, but require a scalable and privacy-conscious way to practise before the examination setting. Examiner Coach is not a replacement for expert-led examiner training, but it offers a possible practice layer between large-group instruction and real OSCE delivery. It also demonstrates how KISSKI-style AI service infrastructure can support an applied health-education scenario, connecting research on AI methods with the practical requirements of sensitive institutional environments.

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